

# The Role of Social Sciences in the Study of Misinformation: A Bibliometric Analysis of Web of Science and Scopus Publications (2017-2022)

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*This study examines the contribution of the Social Sciences to disinformation research. Using network analysis and bibliometrics with the Bibliometrix tool in R, we analyze academic publications in Scopus and Web of Science (WOS) to understand scholarly work on misinformation, disinformation, and fake news. We compare data from Scopus and WOS, explore research trends, identify influential authors, examine relevant journals, assess productive institutions and countries, analyze author keywords, and provide a brief analysis of highly cited articles. The findings reveal the scholarly landscape of disinformation research within the Social Sciences. The comparison of Scopus and WOS data highlights the coverage*

*and representation of disinformation studies. Research trends indicate the field's growth and acceptance through publication and citation rates. Influential authors are identified based on publication output and h-index. Key journals in the field are identified, and productive institutions and countries are assessed. The analysis of author keywords reveals central themes and topics within the discipline. In addition, the analysis of highly cited articles provides insights into the theoretical and methodological aspects that have received significant attention.*

**Keywords:** misinformation, disinformation, fake news, Social Sciences, network analysis, bibliometric.



In recent years, the proliferation of disinformation and fake news has emerged as a critical concern in the digital age. With the rapid growth of online platforms and the ease of information dissemination, distinguishing fact from fiction has become increasingly challenging. The impact of false information on public opinion, political processes, and societal stability necessitates a deep understanding of this phenomenon and effective strategies to counteract its harmful effects.

This study provides a comprehensive analysis of scholarly publications in the field of misinformation, disinformation and fake news studies within the Social Sciences domain. We describe aspects related to scholarly articles on misinformation, disinformation, and fake news within the field of Social Sciences in the global context between 2017 and 2022. We seek to understand the trend of such articles during this time period. Additionally, we aim to identify the prominent and prolific authors who have contributed significantly to the research in this field within Social Sciences. We also investigate which journals play a significant role in publishing research on misinformation, disinformation, and fake news in the same field. Furthermore, we explore the institutions and countries that have been the most productive in scholarly research on the same topics within the identified research scope. We analyze the prevalent author keywords used in the published articles within the specified study framework. Finally, we examine the theoretical and methodological aspects that characterize the most cited articles in this research context.

The methodological approach is based on network analysis and bibliometrics, implemented using the Bibliometrix tool in the R environment and computer language. This approach allows for a comprehensive examination of the scholarly landscape within Social Sciences publications on disinformation and fake news.

The study starts by conducting a systematic literature review, collecting relevant articles from the Scopus and WOS databases. The retrieved data is then subjected to network analysis techniques to analyze collaboration patterns among authors and institutions. Additionally, bibliometric analysis is employed to examine publication rates, citation trends, and the productivity and influence of journals in the field.

The use of network analysis and bibliometrics enables the identification of key authors, emerging thought leaders, and research themes within Social Sciences. The collected data is processed and analyzed using the Bibliometrix tool. The results obtained through this methodological approach contribute to a comprehensive understanding of the field's growth, dynamics, and trends within the Social Sciences domain.

This paper is divided into five sections. The first section gives a brief overview of the study's background, objectives, and research questions. The second section explains the concepts and definitions related to misinformation, disinformation, and fake news that are similar or connected. In the third section, we describe the methodology used for the study, specifically the bibliometric approach. The next section presents the results of our analysis. Finally, we conclude by discussing the limitations of the research and suggesting directions for future studies.

## MISINFORMATION, DISINFORMATION, FAKE NEWS, AND RELATED CONCEPTS

The academic literature on misinformation, disinformation and fake news is vast and recent. The discussion arising from the problems generated by the rapid and vast distribution and exchange of malicious information and messages on the Internet concerns scholars of social sciences, communication, computer science, technology and health. The main concerns are, on the one hand, to define and characterize the phenomenon of communicational and informational disorder, and on the other, to propose ways of detecting and containing its effects.

Although recent, the literature on these three terms indicates that they are not entirely new phenomena. Their definitions have been updated as the media system has undergone changes, mainly caused by the expansion of access to information and content production in the digital environment.

If we go back to the uses of the term fake news by scholars, we see that its definition has changed in recent years (Waisbord, 2018). The expression went from being used to denote humorous and satirical commentaries that imitated the form of conventional news on television programs and in the print media (Chen, 2017; Kiernan, 2017), to refer to content deliberately elaborated with the purpose of misinforming and influencing public opinion. In its current state, the term “refers to fabricated information that astutely mimics news and taps into existing public beliefs to influence electoral behavior” (Waisbord, 2018, p. 1).

Currently, the phenomenon of information disorder, which includes the definitions of misinformation, disinformation and fake news, differs from previous moments in speed, scale and breadth of distribution and consumption of information. Platforms, governments, digital entrepreneurs and consumers are relevant actors in this context (Waisbord, 2018). Social media platforms facilitate the broad presence of content not referenced by traditional media and influence the ways in which the public constitutes social links based on their common interests. Political elites assume a similar role when they appropriate the logics of social networks and messaging applications to provoke confusion in the public and influence public opinion and the construction of political identities. In turn, large digital entrepreneurs plan the financing of their projects from the data of information consumption patterns and disinformation and from the flow of content itself. The common consumer contributes by curating and sharing with their peers.

Misinformation, disinformation and fake news are part of the same phenomenon. In the literature, this phenomenon is called information disorder (Waisbord, 2018; Wardle and Derakhshan, 2017) or perceived untruthfulness (Hameleers *et al.*, 2021, 2022). Both denominations correspond to different perspectives from which scholars propose their view. The first perspective is related to the increasing speed and quantity of information exchanged in the digital environment and the second refers to the perceptions of the veracity and quality of information by users or readers.

Yet misinformation, disinformation and fake news are different from each other. Misinformation or misinformation is unintentional false information

(Hameleers *et al.*, 2021, 2022). In other words, it is the wrong information when evaluated in contrast to the consensus knowledge among experts and the best available evidence (Vraga and Bode, 2020).

Vraga and Bode (2020) develop the definitions offered by Nyhan and Reifler (2010) and Kuklinski *et al.* (2000). Kuklinski and colleagues propose that misinformation occurs when people have inaccurate beliefs and hold them with confidence (Kuklinski *et al.*, 2000), being possible that this happens because of ignorance or lack of knowledge and because of confidence in having knowledge, even if it is inaccurate or wrong. In this case, Kuklinski *et al.* (2000) make the distinction between ignorance and misinformation.

Nyhan and Reifler use this distinction to support the difference between misinformation and misperceptions. As defined by these authors, misperception corresponds to cases in which people's beliefs about the reality of facts are not supported by clear evidence or expert opinion. Thus, misperception has to do with personal beliefs and can be generated by misinformation.

Vraga and Bode (2020) propose a definition of misinformation based on expert consensus and verification of the best available evidence. For the authors, in areas such as health sciences it is more common to build or identify a consensus on a given issue, but in areas such as politics this becomes more complicated. For this, they formulate by virtue of the consideration of the best available evidence as a second parameter of the definition.

The concept of disinformation presupposes the intention to deceive. The definition includes the manipulation, decontextualization and fabrication of targeted information (Tandoc *et al.*, 2018; Wardle, 2017). It is defined as "false information deliberately created to harm a person, social group, organization or country" (Wardle and Derakhshan, 2017, p. 20). Disinformation has many formats and modes of presentation, fake news is a mode of disinformation that imitates news.

The term fake news also goes back centuries and, mainly, in recent decades has acquired another meaning. The study by Tandoc *et al.* (2018) indicates six types of definition for fake news from a review of other studies using the term: (1) news satire, (2) news parody, (3) fabrication, (4) manipulation, (5) advertising, and (6) propaganda. For the authors, a common aspect among these definitions is that "fake news appropriates the look and feel of real news; from how websites look; to how articles are written; to how photos include attributions" (Tandoc *et al.*, 2018, p. 147). Beyond the appearance, fake news act in the construction of networks and connections between bots, blogs, political authorities and public figures that simulate a kind of credibility to the malicious content, seeking to conquer its legitimacy among users and readers.

Beyond the conceptual distinctions between misinformation, disinformation, and fake news, the literature on the subject already gathers relevant evidence to characterize the phenomenon of contemporary information disorder. A significant portion of studies describes conceptual advances and new typologies of disinformation, its modes of creation and circulation, the effects it has on people and institutions, and alternatives for detection and containing its effects.

In a study on the fake news circulating during the SARS-CoV-2 virus pandemic, Salaverría *et al.* (2020) explore the formats, sources, and territories of origin of disinformation to propose a new definition and typological description oriented towards its severity. The authors identify four main types: joke, exaggeration, decontextualization, and deceit. Salaverría's work complements that of Tandoc (2014, 2018, 2019) and Freelon and Wells (2020) in the task of tracking the conceptual developments of fake news.

The discussion about misinformation also covers its modes of creation, distribution, and consumption. Nelson and Taneja (2018) examined data on visits to mobile and desktop platforms in the months leading up to and following the 2016 presidential elections in the United States. Among their findings is that the audience consuming fake news consists of a small, disloyal group of heavy internet users, and social media platforms are responsible for directing traffic to fake news pages. Bakir and McStay (2018) analyze the same period in American history and emphasize the need for advertising regulation to curb the sensational nature of fake news crafted to mobilize users' personal and emotional aspects. It's also worth mentioning the work of Buchanan and Benson (2019), where they highlight the influence of the characteristics of information sources and recipient on the organic spread of misinformation.

The years encompassed by this study are marked by the SARS-CoV-2 pandemic, and as such, many studies on the effects of misinformation are related to public health (Kim *et al.*, 2020). This is the case with the work "Measuring the impact of COVID-19 vaccine misinformation on vaccination intent in the UK and USA", by Loomba *et al.* (2021). The authors investigate to what extent exposure to misinformation affects the vaccination intent of the populations in the USA and the UK. The results indicate a decrease in vaccination intent in both countries, and also that certain sociodemographic groups are affected differently when exposed to vaccine misinformation. Additionally, they observe that misinformation mimicking scientific discourse is more strongly associated with a decline in vaccination intent. Equally relevant, other works also report the effects of misinformation on the political and media systems of different countries (Vaccari and Chadwick, 2020; Duyn and Collier, 2019; Vargo *et al.*, 2018; Zimmermann and Kohring, 2020).

There is also a growing concern about the deliberate and strategic spread of misinformation. In recent years, misinformation has been one of the contributing factors to political polarization, the dissemination of conspiracy theories, verbal aggression towards public figures, and the distrust of democratic institutions such as government structures and the press. As a result, the number of studies aiming to investigate and propose ways to detect and combat misinformation and its effects is increasing. We highlight the meta-analytical studies by Nathan Walter and his colleagues on effective forms of information correction (Walter and Tukachinsky 2020; Walter and Murphy, 2018; Walter and Tukachinsky, 2020). It is also worth mentioning research on the efficiency of fact-checking (Graves, 2018; Hameleers and Meer, 2020) and studies on the role of government officials in promoting distrust in health policies and science through the use of misinformation (Lovari, 2020).

## METHODOLOGY AND BIBLIOMETRIC ANALYSIS IN THE SOCIAL SCIENCES

In recent years, the number of scientific publications in all areas of knowledge using bibliometric analysis as a method has increased (Donthu *et al.*, 2021). Although the first bibliometric analyses date back to the 1950s, their use in the Social Sciences is recent. More than accompanying the growth of the scientific field itself, this trend is also related to the emergence of scientific databases such as Scopus and WOS, which are capable of collecting and organizing large volumes of bibliometric data. In addition, the development of software that allows and facilitates the reading and organization of bibliometric research results has sparked the interest of scholars.

By definition, bibliometric analysis is essentially quantitative and descriptive in nature. The technique is used by researchers with the aim of discovering emerging trends in the performance of articles and publications, patterns of collaboration between member centers, groups, research institutions and universities, and also to explore the intellectual structure of a specific field in the existing literature (Donthu *et al.*, 2021).

According to Donthu *et al.* (2021), bibliometric analysis techniques fall into two categories: (1) performance analysis and (2) scientific mapping. Briefly defined, the first focuses on the contributions of the components of the research field, and the second is centered on the relationships generated between them. In our analysis, both approaches are of interest. Our objective is to understand the contribution of the Social Sciences to the production of scientific knowledge on misinformation in recent years by analyzing the scientific production and the contribution and thematic networks between authors and institutions.

Even with the growing academic interest and vast literature on misinformation, bibliometric analysis on this topic is scarce. To mention a few studies, Obreja (2022) conducted research on how the political emphasis on social media has evolved in scholarly publications published in the last decade. Patra *et al.* (2022) investigated the scholarly contribution of published literature on the term fake news and associated terms such as misinformation, disinformation, and post-truth in various disciplines. In turn, Sahoo *et al.* (2021) made a quantitative and qualitative assessment of published research on fake news in light of the Library and Information Science perspective and showed research trends using bibliometrics.

Table 1. Data collection process

|                    | Scopus <sup>1</sup>                                                                                       | Web of Science (WOS) <sup>2</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Main search terms  | Misinformation, disinformation, and fake news                                                             | Misinformation, disinformation, and fake news                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Selection criteria | Document Type: Article<br>Publication Stage: Final<br>Time Periods: 2017-2022<br>Subarea: Social Sciences | Document Type: Article<br>Time Periods: 2017-2022<br>WOS Index: Social Sciences Citation Index (SSCI)<br>Research Areas: Communication or Psychology or Government Law or Social Sciences Other Topics or Sociology or International Relations or Biomedical Social Sciences or History Philosophy of Science or Philosophy or Social Issues or Cultural Studies or History or Public Administration or Religion or Ethnic Studies or Anthropology or Behavioral Sciences or Family Studies or Urban Studies |
| N                  | 1821                                                                                                      | 1742                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Source: Prepared by the author.

1 Scopus search: TITLE-ABS-KEY (misinformation OR disinformation OR fake AND news) AND (LIMIT-TO (PUBSTAGE , “final”)) AND (LIMIT-TO ( DOCTYPE, “ar”)) AND (LIMIT-TO (SUBJAREA , “SOCI”)) AND (LIMIT-TO (LANGUAGE , “English”)) AND (LIMIT-TO (SRCTYPE, “j”)) AND (LIMIT-TO (PUBYEAR, 2022) OR LIMIT-TO (PUBYEAR , 2021) OR LIMIT-TO (PUBYEAR , 2020) OR LIMIT-TO (PUBYEAR , 2019) OR LIMIT-TO (PUBYEAR , 2018) OR LIMIT-TO (PUBYEAR , 2017))

2 WOS search: ALL=(misinformation OR disinformation OR fake news) and Article (Document Types) and 2022 or 2021 or 2020 or 2019 or 2018 or 2017 (Publication Years) and English (Languages) and Social Sciences Citation Index (SSCI) (Web of Science Index) and Communication or Psychology or Government Law or Social Sciences Other Topics or Sociology or International Relations or Biomedical Social Sciences or History Philosophy of Science or Philosophy or Social Issues or Cultural Studies or History or Public Administration or Religion or Ethnic Studies or Anthropology or Behavioral Sciences or Family Studies or Urban Studies (Research Areas) and Early Access or Book Chapters or Proceeding Paper (Exclude – Document Types) and Ergonomics or Gerontology or Literature or Respiratory System or Sport Sciences or Telecommunications or Ecology or Veterinary Sciences or Rehabilitation or Pharmacology Pharmacy or Nursing or Mathematics Interdisciplinary Applications or Genetics Heredity or Food Science Technology or Endocrinology Metabolism or Architecture or Zoology or Transportation or Oncology or Medicine Legal or Engineering Industrial or Development Studies or Language Linguistics or Environmental Studies or Psychology Educational or Economics or Regional Urban Planning or Medical Ethics or Area Studies or Hospitality Leisure Sport Tourism or Geography or Psychology Mathematical or Physiology or Pediatrics or Substance Abuse or Geriatrics Gerontology or Nutrition Dietetics or Engineering Multidisciplinary or Obstetrics Gynecology or Social Work or Computer Science Artificial Intelligence or Public Environmental Occupational Health or Health Policy Services or Education Educational Research (Exclude – Web of Science Categories).

Our research aims to describe the contribution of Social Sciences in the study of misinformation using the performance analysis and scientific mapping techniques of bibliometrics. Therefore, we propose the following research questions:

- Research question 1: What is the trend of Social Science studies published as articles on misinformation, disinformation and fake news in recent years (2017-2022) in the world?
- Research question 2: In this same field of study, who are the most relevant and productive authors?
- Research question 3: Which are the most relevant journals?
- Research question 4: In the same research logic, which are the most productive institutions and countries?
- Research question 5: What are the keywords most used by the authors in the articles published according to this same research proposal?
- Research question 6: What theoretical and methodological aspects characterize the most cited articles?

In data collection, we collected publications from the two most reputable academic citation databases: Scopus and WOS. Both databases are considered the main sources of bibliographic data and indexing of scientific articles and are widely used in bibliometric analysis studies. (Echchakoui, 2020) We set up a search strategy including the terms associated with the phenomenon of information disturbance such as misinformation, disinformation, and fake news, with some conditions to retrieve the bibliographic data from the approach we propose in the Social Sciences. Table 1 shows all the criteria we used for the search in the databases. The search was carried out with the words in English to cover a more expressive number of publications. We also limited the search to articles (papers), Social Sciences (Social Sciences) and the years between 2017 and 2022 —these are the years in which there was an expressive increase in publications (Patra *et al.*, 2022). After examining the two databases independently, we manually reviewed the authors' names, titles, and abstract, excluding articles whose topics were unrelated to our study.

As a result, we identified 1821 publications in Scopus and 1742 in WOS. The search was conducted on April 14, 2023. For data analysis, table and graph preparation, we utilized the R library, Bibliometrix (Aria and Cuccurullo, 2017). Our preference for Bibliometrix stemmed from its comprehensive suite of functions designed for bibliometric analysis, enabling an in-depth exploration of scientific collaboration, research trends, and performance metrics. Furthermore, Bibliometrix's open-source nature enhances transparency and reproducibility in our methodology. While user-friendly and visually appealing, VOSviewer's preprogrammed functions may limit flexibility. In contrast, Bibliometrix, operating within the R software environment, allows greater customization and supports comprehensive mapping analysis.

The data analysis was performed according to the following steps: (a) a general comparison of Scopus and WOS data, (b) a description of research trends, to



understand the progression of the field and its acceptance through the number of publications and citations per year, (c) a description of the most relevant authors, according to the number of publications and h-index, (d) a description of the most relevant journals, (e) a description of the most productive institutions and countries, (f) a description of the keywords most used by the authors, and (g) a brief analysis of the most cited articles.

RESULTS

The analysis of scientific articles in Social Sciences published between 2017 and 2022 includes 1821 documents in Scopus and 1742 documents in WOS. In both databases, we identify a quite significant annual growth percentage of the number of documents, observing Table 2 we see that in Scopus the growth is 55.18% and in WOS is 44.12%. The same table shows that the publications are distributed among 752 journals in the case of Scopus and 389 in WOS. The total number of authors is similar in the two databases: 3843 in Scopus and 3808 in WOS. In Scopus 580 are single authors of articles and in WOS there are 418, contributing to a higher rate of co-authors per document in WOS (= 2.85) than in Scopus (= 2.44). WOS also surpasses Scopus in terms of international cooperation among authors, having 24.28% of international co-authorships of articles while Scopus has 16.91%.

Table 2. General database information

| Description                     | Scopus    | Web of Science (WOS) |
|---------------------------------|-----------|----------------------|
| Timespan                        | 2017:2022 | 2017:2022            |
| Sources (journals, books, etc.) | 752       | 389                  |
| Documents                       | 1821      | 1742                 |
| Annual Growth Rate %            | 55.18     | 44.12                |
| Document Average Age            | 2.63      | 2.53                 |
| Average citations per doc       | 14.13     | 18.32                |
| <b>DOCUMENT CONTENTS</b>        |           |                      |
| Author's Keywords (DE)          | 4315      | 3852                 |
| <b>AUTHORS</b>                  |           |                      |
| Authors                         | 3843      | 3808                 |
| Authors of single-authored docs | 580       | 418                  |
| <b>AUTHORS COLLABORATION</b>    |           |                      |
| Single-authored docs            | 622       | 452                  |
| Co-authors per Doc              | 2.44      | 2.85                 |
| International co-authorships %  | 16.91     | 24.28                |

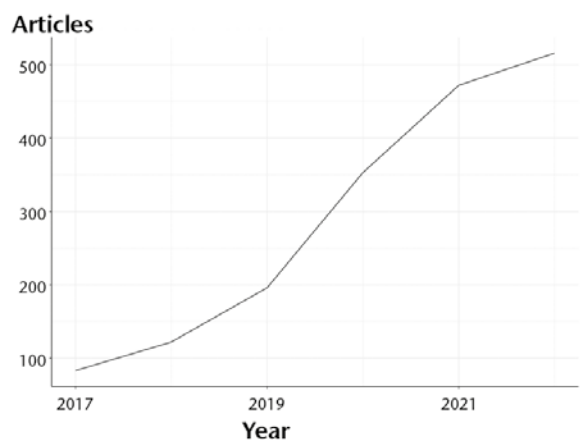
Source: Prepared by the author.

RESEARCH TRENDS

The growth trend of disinformation and fake news studies is identified in bibliometric analysis studies. In similar work comprising publications on fake

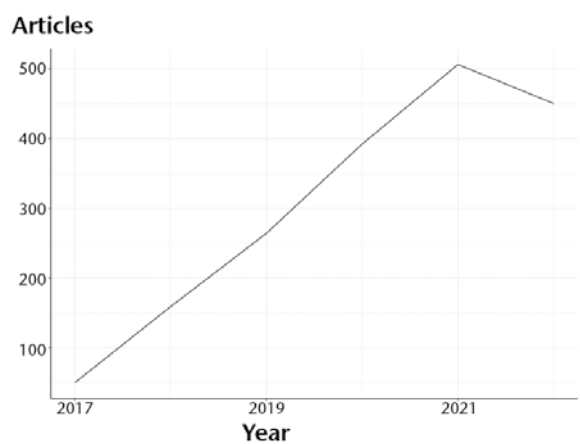
news in Scopus and WOS between the years 2001 and 2020, Patra *et al.* identified that the “annual publication trend almost doubled from 2017 in both databases. Annual citations gradually increased in both databases” (Patra *et al.*, 2022, p. 6). In Figures 1 and 2, our work indicates similar trend, except for Scopus that presents a retraction in 2022. In Scopus (Figure 2), the annual scientific production corresponds to 50 articles in 2017, 159 in 2018, 264 in 2019, 392 in 2020, 506 in 2021 and a decrease to 450 in 2022. In WOS (Figure 1), we have 83 articles in 2017, 122 in 2018, 196 in 2019, 353 in 2020, 472 in 2021, and 516 in 2022.

**Figure 1. Annual scientific production in Web of Science (WOS).  
Frequency of articles published per year (2017-2022)**



Source: Elaborated by the author using Bibliometrix.

**Figure 2. Annual scientific production in Scopus.  
Frequency of articles published per year (2017-2022)**



Source: Elaborated by the author using Bibliometrix.

In Table 3, we present a comparison of the average annual citations in WOS and Scopus. Although the numbers do not coincide between the two, we note that the trend is similar when we consider the movement of increase and decrease in each of the years studied. That is, in both databases, the average number of citations per year, increases from 2017 to 2018, decreases from 2018 to 2019, increases again from 2019 to 2020, decreases from 2020 to 2021 and decreases further from 2021 to 2022. It is important to consider that the 2021 and 2022 data tend to change over the years as citations to newly published articles increase.

**Table 3. Average number of citations per year in Web of Science (WOS) and Scopus**

| WOS  |         | Scopus |         |
|------|---------|--------|---------|
| Year | Average | Year   | Average |
| 2017 | 6,55    | 2017   | 3,89    |
| 2018 | 8,41    | 2018   | 5,32    |
| 2019 | 5,07    | 2019   | 4,14    |
| 2020 | 6,72    | 2020   | 4,86    |
| 2021 | 4,02    | 2021   | 3,25    |
| 2022 | 1,74    | 2022   | 1,41    |

Source: Prepared by the author.

**COMPARISON OF THE NUMBER OF PUBLICATIONS AND CITATIONS BY AUTHORS**

Table 4 shows the ranking of the top ten authors in both databases ordered according to the h-index. Both databases coincide in listing among the top ten the names of Tandoc, Vraga, Hameleers, Bode and Van der Linden. However, in WOS the presence of Lewandowski, Ecker, Pennycook, Rand and Otgaar contrasts; in Scopus, Tully, Roozenbeek, Wang, Duffy and Lee.

Total publications (NP) and total citations indicate different positions among authors. According to the number of publications (NP), in WOS, Ecker, Lewandowsky, Otgaar (= 21) followed by Hameleers (= 18). In Scopus, Hameleers (= 13), Tandoc and Wang (= 9) and Tully (= 8) occupy the first positions. By total citations (TC), we highlight in WOS the names of Pennycook (= 2279) and Rand (2252), which present approximate numbers compared to the others in the same database. In Scopus, we note approximate numbers between Vraga (= 590), Roozenbeek (= 538), Van der Linden (= 486) and Bode (= 465).

Table 4. Local impact of authors, ordered according to the h-index

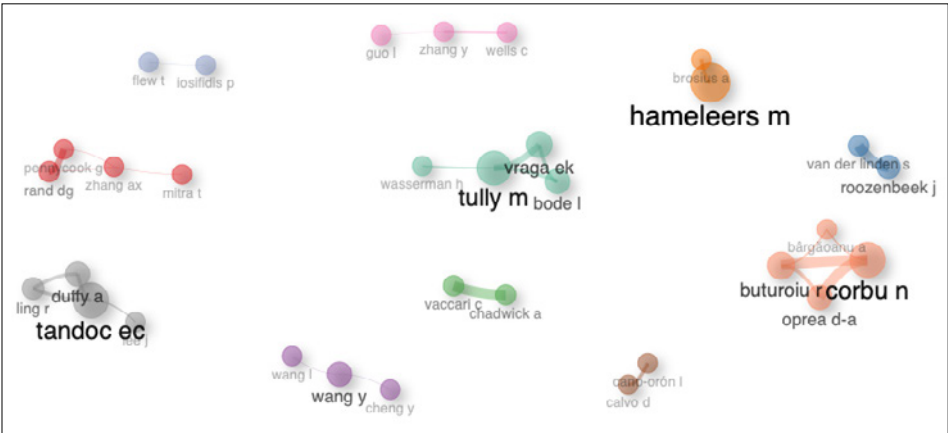
| Web of Science (WOS) |    |      |         | Scopus           |    |     |         |
|----------------------|----|------|---------|------------------|----|-----|---------|
| Author               | NP | TC   | h-index | Author           | NP | TC  | h-index |
| LEWANDOWSKY S        | 21 | 1349 | 14      | TANDOC EC        | 9  | 413 | 7       |
| ECKER UKH            | 21 | 1169 | 12      | TULLY M          | 8  | 277 | 7       |
| VRAGA EK             | 12 | 620  | 12      | HAMELEERS M      | 13 | 226 | 6       |
| PENNYCOOK G          | 14 | 2279 | 10      | ROOZENBEEK J     | 6  | 538 | 6       |
| BODE L               | 10 | 459  | 9       | VRAGA EK         | 6  | 590 | 6       |
| HAMELEERS M          | 18 | 286  | 9       | WANG Y           | 9  | 95  | 6       |
| RAND DG              | 14 | 2252 | 9       | BODE L           | 6  | 465 | 5       |
| TANDOC EC            | 12 | 1164 | 9       | DUFFY A          | 6  | 284 | 5       |
| VAN DER LINDEN S     | 10 | 528  | 9       | LEE J            | 6  | 66  | 5       |
| OTGAAR H             | 21 | 161  | 8       | VAN DER LINDEN S | 5  | 486 | 5       |

Source: Prepared by the author. NP: number of publications. TC: total citations.

Network analysis contributes to deepen the description of the numbers presented above. In this case, the data referring to the collaboration networks between authors indicate correlation with the productivity indexes (NP and TC). In Figures 3 and 4 we see how the authors are associated. Among the authors mentioned in the Scopus list, we see that the authors are distributed in the following clusters (Figure 3):

- Blue cluster: Roozenbeek and Van der Linden;
- Gray cluster: Tandoc, Duffy and Lee;
- Light blue cluster: Tully, Bode and Vraga;
- Orange cluster: Hameleers and Brosius;
- Purple cluster: Y Wang with other authors.

Figure 3. Network of collaboration between authors in Scopus

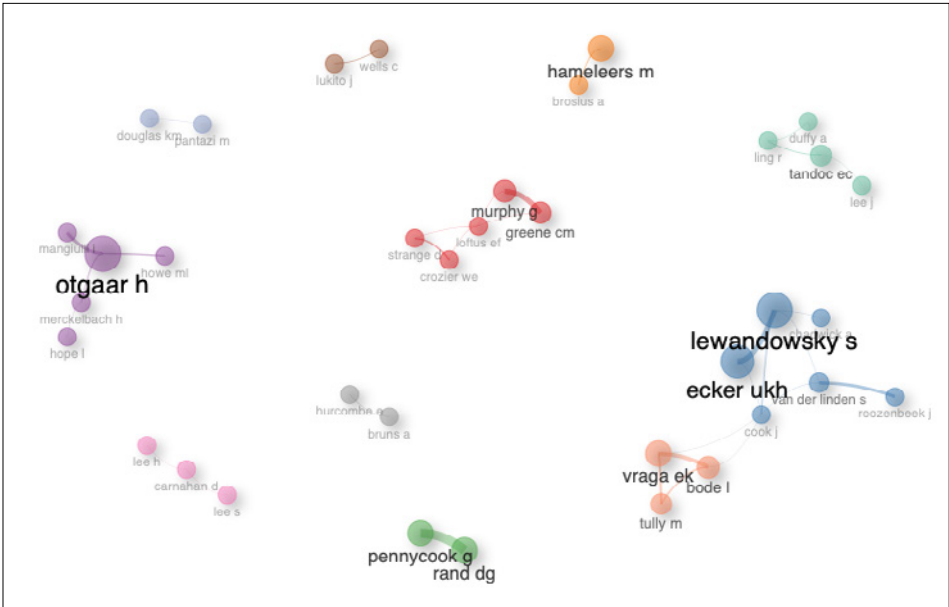


Source: Elaborated by the author using Bibliometrix.

Among the authors mentioned in the WOS list, we noted the following distribution of clusters (Figure 4):

- Blue cluster: associates Lewandowsky, Ecker and Van der Linden;
- Light blue cluster: associates Tandoc with Duffy, Lee and Ling who are not listed in Table 4;
- Light red cluster: Tully, Bode and Vraga;
- Green cluster: Pennycook and Rand;
- Orange cluster: there are Hamелеers and Brosius.
- Purple cluster: there is Otgaar with other authors.

Figure 4. Network of collaboration between authors in Web of Science (WOS)



Source: Elaborated by the author using Bibliometrix.

Thus, in both databases, the most productive authors are associated with other authors in similar clusters. The five clusters we identified form five distinct collaborative networks with the same authors in WOS and Scopus.

COMPARISON OF PUBLICATIONS AND CITATIONS BY JOURNAL

Our study only considered articles published in journals. Thus, Tables 5 and 6 present the ten most productive and influential journals in WOS and Scopus. In WOS, the journal *Social Media + Society* presents the highest total number of publications (NP) with 75 articles, followed by *Frontiers in Psychology* with 51 publications. According to the same database, *Digital Journalism* has the highest total number of citations with 1828 and is followed by *New Media and Society* with 1218.

**Table 5. Local impact of publications in Web of Science (WOS), ordered according to the h-index**

| Journal                                             | NP | TC   | h-index |
|-----------------------------------------------------|----|------|---------|
| New Media and Society                               | 31 | 1218 | 18      |
| Social Media + Society                              | 75 | 1001 | 18      |
| Digital Journalism                                  | 48 | 1828 | 16      |
| Political Communication                             | 19 | 849  | 15      |
| Information Communication and Society               | 27 | 745  | 14      |
| Computers in Human Behavior                         | 27 | 721  | 12      |
| Frontiers in Psychology                             | 51 | 841  | 12      |
| International Journal of Press-Politics             | 15 | 394  | 12      |
| Journal of Applied Research in Memory and Cognition | 27 | 1116 | 12      |
| Psychological Science                               | 14 | 1195 | 12      |

Source: Prepared by the author. NP: number of publications. TC: total citations.

In Scopus (Table 6), the highest total number of publications (NP) and total citations (TC) is *Digital Journalism* with 37 articles and 1107 citations. This is followed by *Social Media and Society* and *Media and Communication* with 35 publications. As for the total number of citations, *New Media and Society* has 1064.

**Table 6. Local impact of publications in Scopus, ordered according to the h-index**

| Journal                                              | NP | TC   | h-index |
|------------------------------------------------------|----|------|---------|
| Social Media and Society                             | 35 | 761  | 16      |
| Digital Journalism                                   | 37 | 1107 | 14      |
| Information Processing and Management                | 20 | 935  | 14      |
| New Media and Society                                | 21 | 1064 | 13      |
| Information Communication and Society                | 15 | 530  | 10      |
| Journalism Studies                                   | 25 | 701  | 10      |
| Media and Communication                              | 35 | 435  | 10      |
| Proceedings of the ACM on Human-Computer Interaction | 25 | 275  | 10      |
| American Behavioral Scientist                        | 12 | 693  | 9       |
| Journalism Practice                                  | 16 | 243  | 9       |

Source: Prepared by the author. NP: number of publications. TC: total citations.

**MOST PRODUCTIVE INSTITUTIONS AND COUNTRIES**

In Table 7 we list the most productive institutions according to the number of publications. According to WOS data, the University of Oxford (= 61), the University of Wisconsin (= 43) and the University of Amsterdam (= 40) occupy the top three places in the list. In Scopus, the top of the list is the University of Amsterdam (= 50), which is followed by the University of Texas (= 46) and the University of California (= 33). The only institutions that are in both databases

are University of Oxford, University of Amsterdam, University of Texas and Nanyang Technological University.

**Table 7. Most productive institutions according to the number of publications**

| WOS                              |    | Scopus                                                         |    |
|----------------------------------|----|----------------------------------------------------------------|----|
| Institution                      | NP | Institution                                                    | NP |
| University Oxford                | 61 | University of Amsterdam                                        | 50 |
| University Wisconsin             | 43 | University of Texas at Austin                                  | 46 |
| University Amsterdam             | 40 | University of California                                       | 33 |
| University Texas Austin          | 40 | Nanyang Technological University                               | 31 |
| University Western Australia     | 36 | University of Oxford                                           | 29 |
| Maastricht University            | 35 | University of New Mexico                                       | 28 |
| Nanyang Technological University | 35 | Nat. University of Political Studies and Public Administration | 25 |
| Harvard University               | 34 | Queensland University of Technology                            | 21 |
| Northwestern University          | 30 | Boston University                                              | 20 |
| University Michigan              | 28 | Dartmouth College                                              | 20 |

Source: Prepared by the author. NP: number of publications.

The analysis by country (Table 8) took into account the affiliations of academics in the documents. In both databases, there is the predominance of the United States with the frequency of 1637 author affiliations in WOS and 1282 in Scopus. The difference between the countries listed in the databases is the presence of Italy and Singapore in WOS and Brazil and India in Scopus. The other countries are the same, changing only the order in which they appear in the lists.

**Table 8. Most productive countries, considering the frequency of affiliation of scholars in the documents**

| WOS         |       | Scopus      |       |
|-------------|-------|-------------|-------|
| Country     | Freq. | Country     | Freq. |
| USA         | 1637  | USA         | 1282  |
| UK          | 421   | Spain       | 293   |
| Australia   | 215   | UK          | 272   |
| Germany     | 146   | Australia   | 175   |
| Netherlands | 146   | India       | 174   |
| Canada      | 132   | Brazil      | 140   |
| China       | 122   | Canada      | 129   |
| Spain       | 110   | Germany     | 120   |
| Italy       | 84    | China       | 118   |
| Singapore   | 61    | Netherlands | 95    |

Source: Prepared by the author.

**MOST FREQUENT AUTHOR KEYWORDS AND CO-OCCURRENCE ANALYSIS**

In Table 9 we show the most frequent author keywords among the analyzed papers (2017-2022) in the two databases. At the top of the list are “misinformation”

in WOS with 454 mentions and “fake news” in Scopus with 778. The first seven words in the two databases are the same. The difference is that some of these words occupy different positions in the ranking, but they are still “misinformation”, “fake news”, “social media”, “disinformation”, “covid-19”, “twitter” and “journalism”.

On March 11, 2020, the World Health Organization declared a pandemic situation for the Sars-Cov-2 virus. Among the pandemic-related words, we found “covid-19” and “coronavirus” in WOS; in Scopus are “covid-19”, “coronavirus” and “pandemic”.

**Table 9. Most frequent author keywords among the analyzed papers (2017-2022) in the two databases**

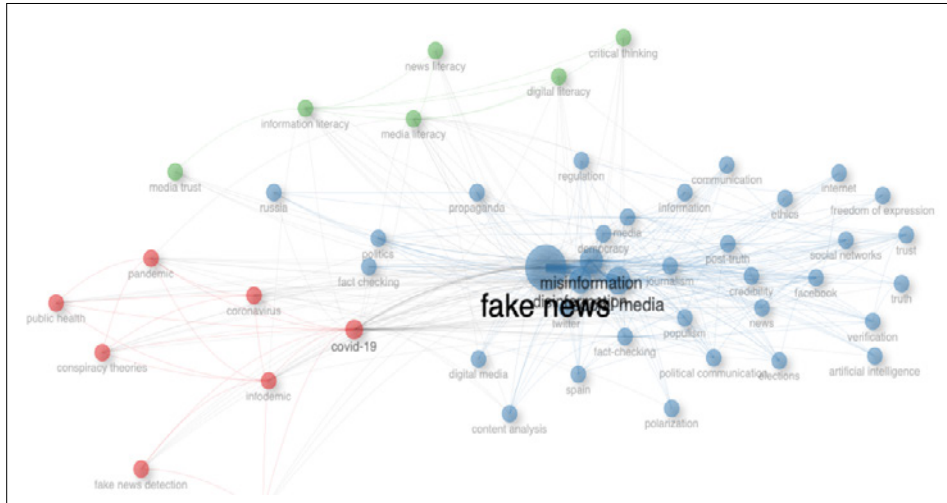
| WOS                     |       | Scopus               |       |
|-------------------------|-------|----------------------|-------|
| Author keywords         | Freq. | Author keywords      | Freq. |
| misinformation          | 454   | fake news            | 778   |
| fake news               | 344   | social media         | 357   |
| social media            | 305   | misinformation       | 304   |
| disinformation          | 253   | disinformation       | 295   |
| covid-19                | 172   | covid-19             | 211   |
| twitter                 | 64    | journalism           | 84    |
| journalism              | 63    | twitter              | 77    |
| false memory            | 54    | media literacy       | 72    |
| facebook                | 51    | post-truth           | 68    |
| fact-checking           | 50    | fact-checking        | 64    |
| memory                  | 43    | media                | 59    |
| conspiracy theories     | 41    | information literacy | 51    |
| propaganda              | 38    | facebook             | 49    |
| political communication | 32    | credibility          | 45    |
| elections               | 31    | infodemic            | 45    |
| trust                   | 30    | trust                | 44    |
| coronavirus             | 29    | propaganda           | 42    |
| media                   | 29    | democracy            | 39    |
| politics                | 29    | coronavirus          | 37    |
| populism                | 28    | pandemic             | 35    |

Source: Prepared by the author. NP: number of publications.

In Figures 5 and 6, we highlight the co-occurrence network of authors’ keywords formed by association between Scopus and WOS data, respectively. “The aim of the co-word analysis is to draw the conceptual structure of a framework using a word co-occurrence network to map and cluster terms extracted from keywords, titles, or abstracts in a bibliographic collection.” (Aria and Cuccurullo, 2017, p. 969) In the graphs of the two databases, we notice the predominance of three main clusters. In Scopus, we have a cluster in which the main term is “fake news”, another cluster in which the term is “covid-19” and a last one that has “media literacy”. In WOS, the three main groups are defined by the terms “fake news”, “misinformation” and “covid-19”.

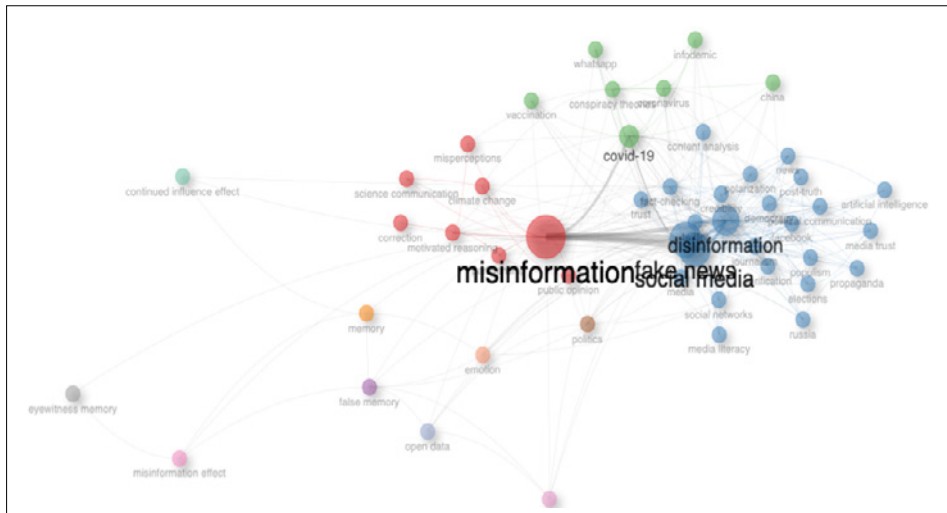


**Figure 5. Keyword co-occurrence network in Scopus**



Source: Elaborated by the author using Bibliometrix.

**Figure 6. Keyword co-occurrence network in Web of Science (WOS)**



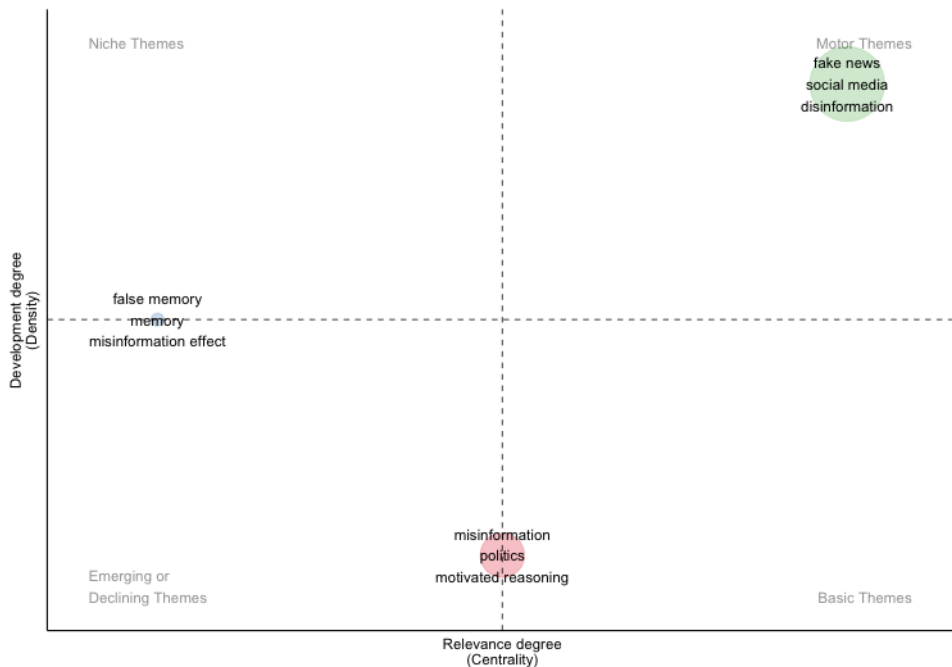
Source: Elaborated by the author using Bibliometrix.

The analysis of the thematic map of the authors' keywords indicates the strength of association between the words by measures of centrality and density. It also proposes a way of visualizing this relationship by a strategic diagram that is divided into four quadrants (Aria *et al.*, 2022). Each quadrant corresponds to a type of theme:

- Motor Themes or Hot Topics: with higher values of centrality and density, they define the relevant words to structure the conceptual framework of the field of study;
- Basic Themes or Topics: with higher values of centrality and lower values of density, they define the topics that are significant to the field of study and transversal to its different areas;
- Niche Themes or Topics: with the lowest values of centrality and the highest values of density, they define topics that are highly developed but marginal to the field of research;
- Emerging or Developing Themes or Peripheral Topics: with the lowest values of centrality and density, they define topics that are not fully developed and are marginally interesting to the field.

Figures 7 and 8 correspond to the strategic diagrams of the authors' keywords according to the WOS and Scopus databases, respectively. In the WOS thematic map, we highlight “fake news”, “social media” and “disinformation” as driving themes. The other two groups of words that are strongly associated with each other are on the borderline between emerging topics and basic and niche topics.

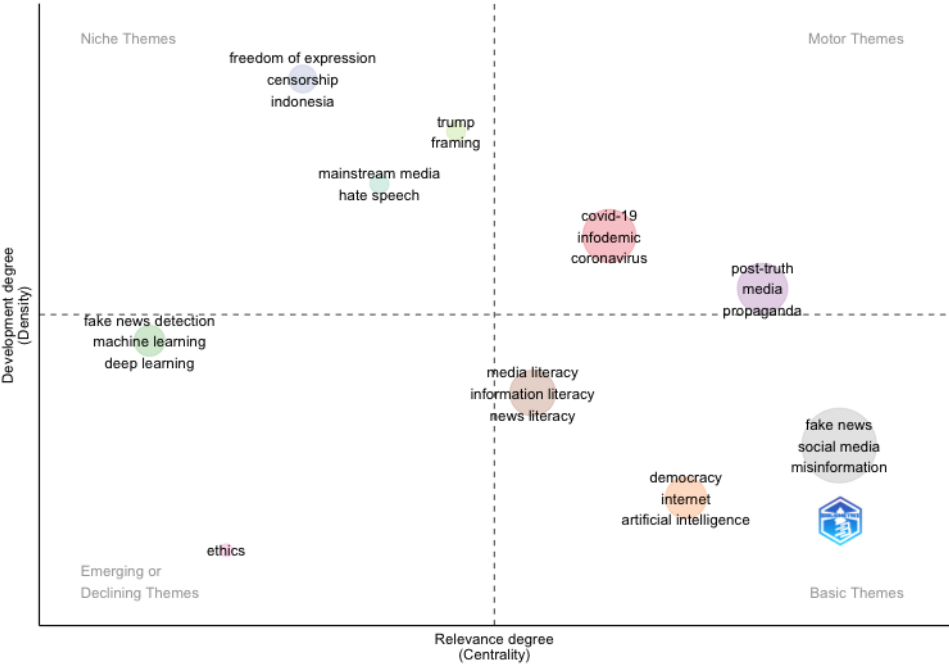
**Figure 7. Thematic map of authors' keywords in Web of Science (WOS)**



Source: Elaborated by the author using Bibliometrix.

The thematic map of author keywords elaborated according to Scopus data shows more clusters of association. In the quadrant of driving themes, there are two distinct clusters: (1) with the terms “covid-19”, “infodemic” and “coronavirus”; (2) with “post-truth”, “media” and “propaganda”. Among the basic topics, there are three clusters: (1) with “fake news”, “social media” and “misinformation”; (2) with “democracy”, “internet” and “artificial intelligence”; (3) with “media literacy”, “information literacy” and “news literacy”. In the niche topics, we also found three groups of words: (1) with “freedom of expression”, “censorship” and “indonesia”; (2) with “trump” and “framing”; (3) with “mainstream media” and “hate speech”. In the emerging topics, we also found three groups of words: (1) with “freedom of expression”, “censorship” and “indonesia”; (2) with “trump” and “framing”; (3) with “mainstream media” and “hate speech”. In the emerging topics, there are two clusters: (1) with the terms “fake news detection”, “machine learning” and “deep learning”; (2) with “ethics”.

Figure 8. Thematic map of authors’ keywords in Scopus



Source: Elaborated by the author using Bibliometrix.

### APPROACHES OF THE MOST CITED ARTICLES

The list with the ten most cited papers between 2017 and 2022 compiles very different articles comparing the two databases. Only two texts are present in the two lists. They are: “Lazy, not biased: Susceptibility to partisan fake news is better explained by lack of reasoning than by motivated reasoning”, by Gordon Pennycook and David Rand (2019); and “The disinformation order: Disruptive

communication and the decline of democratic institutions”, by W. Lance Bennett and Steven Livingston (2018). At the top of the list of most cited articles between 2017 and 2022 in WOS and Scopus are the papers by Edson Tandoc, Zheng Wei Lim and Richard Ling (2018) with 765 citations and that of Gordon Pennycook and David Rand (2019) with 659 citations, respectively.

The article “Defining ‘Fake News’”, by Tandoc, Lim and Ling is published in the journal *Digital Journalism*. The text makes a review of the definitions elaborated by other scholars for the term “Fake news”. According to these definitions, it proposes a typology for the term fake news according to the parameters of factuality and intention. The text results from analysis of 34 academic articles published between 2003 and 2017.

In turn, authors Pennycook and Rand appear with three articles among the ten most cited in WOS. They are, in descending order of CT: “Fighting COVID-19 Misinformation on Social Media: Experimental Evidence for a Scalable Accuracy-Nudge Intervention” (2020) with the contribution of other authors; the previously mentioned “Lazy, not biased: Susceptibility to partisan fake news is better explained by lack of reasoning than by motivated reasoning” (2019), and “Prior Exposure Increases Perceived Accuracy of Fake News” (2018) with the contribution of Tyrone Cannon. All three studies aim to understand why and how people share and believe fake news. Very briefly, the authors are devoted to understanding the cognitive and psychological causes and effects of misinformation on people.

**Table 10. Top 10 articles with the most citations in Web of Science (WOS) and Scopus (2017-2022)**

| WOS                                    |     | Scopus                    |     |
|----------------------------------------|-----|---------------------------|-----|
| Paper                                  | TC  | Paper                     | TC  |
| Tandoc E. C. (2018)                    | 765 | Pennycook G. (2019)       | 659 |
| Van Dam N. T. (2018)                   | 616 | Romer D. (2020)           | 466 |
| Pennycook G. (2020)Pennycook G. (2019) | 607 | Bennett W. L. (2018)      | 463 |
| Loomba S. (2021)                       | 562 | Bakir V. (2018)           | 372 |
| Lewandowsky S. (2017)                  | 536 | Zhang X. (2020)           | 330 |
| Bennett W. L. (2018)                   | 480 | Bode L. (2018)            | 310 |
| Allington D. (2021)                    | 397 | Vargo C. J. (2018)        | 289 |
| Flynn D. J. (2017)                     | 378 | Waisbord S. (2018)        | 275 |
| Pennycook G. (2018)                    | 374 | Laato S. (2020)           | 242 |
|                                        | 372 | Casero-Ripollés A. (2020) | 241 |

Source: Prepared by the author. TC: total citations.

DISCUSSION

The comprehensive analysis of scientific articles in Social Sciences published between 2017 and 2022 has provided insights into the trends, productivity, and impact of research on disinformation and fake news. Notably, our research

trends align with the observations made by Parra Valero and Oliveira (2018), Wang (2020), Sahoo *et al.* (2021), Patra *et al.* (2022), Obreja (2022) and Akram *et al.* (2022), reflecting a significant increase in annual publications and citations. The discrepancy in Scopus's retraction in 2022, unlike WOS, prompts further investigation into the factors influencing publication dynamics. The results of our study indicate an effort of the Social Sciences to understand the phenomena related to misinformation. We identified the growth trend of publications on the subject in WOS and Scopus databases between 2017 and 2022. The annual growth percentage of the number of documents in Scopus is 55.18% and in WOS is 44.12% in the same period.

Among the most relevant authors, there are Lewandowsky in WOS and Tandoc in Scopus, if we consider the h-index. Tandoc, Vraga, Hameleers, Bode and Van der Linden are present in the list of the ten most impacting authors in both databases. Comparing to the results of Patra *et al.* (2022), we note that the names of Ecker, Lewandowsky, Vraga, Tandoc, Van der Linden, Rand, Hameleers, and Roozenbeek are recurrent, indicating a strong impact of these researchers in this field of study. We also note that the most impactful and productive authors are part of collaborative networks with each other.

The affiliation of the authors responds to the question about the most productive country in which the United States predominates with a wide difference in comparison with the others according to both databases as reported in other studies as well (Akram *et al.*, 2022; Patra *et al.*, 2022; Wang, 2020). The top of the lists with the most productive universities is occupied by the University of Oxford and the University of Amsterdam, in WOS and Scopus, respectively. We consider it important to mention also the presence of the University of Texas and Nanyang Technological University in both databases along with those mentioned above. In the work published by Patra *et al.* (2022), the most productive institutions also include Harvard University, Nanyang Technological University, Boston University, University of Oxford, University of Texas Austin, University of California, University of Amsterdam, and Boston University.

The analysis of the authors' keywords shows that, in publications in the same time period, in the two databases, the terms chosen by the authors to define their work are similar. The words "misinformation", "fake news", "social media", "disinformation", "covid-19", "twitter" and "journalism" are the most frequent, differing only in the order in which they appear. The same result appears in the systematic reviews conducted by Akram *et al.* (2022) and Patra *et al.* (2022). They only differ from our results in terms of the significance that COVID-19 comes to assume in the years following 2020 and the presence of the word "Post-Truth" in the systematic review made by Patra *et al.*

Finally, in WOS the most cited text is "Defining 'Fake News'", by Tandoc, Lim and Ling, which is published in *Digital Journalism*. In Scopus, the text "Lazy, not biased: Susceptibility to partisan fake news is better explained by lack of reasoning than by motivated reasoning", by Gordon Pennycook and David Rand, is the most cited. Regarding journals with the highest number of published articles and impact in the field, when we compare the results of our study with those of Akram *et al.* (2022) and Patra *et al.* (2022), we observe that the journals *Social*

*Media + Society, Information, Communication and Society, New Media and Society, and Digital Journalism* are common points among the most relevant in the field.

## CONCLUSION

This article provides an overview of Social Sciences studies on misinformation and related concepts. We show the trend of research in recent years, the main actors among scholars, publications and institutions and the collaboration they establish with each other. More research on the subject is needed, mainly to deepen the disciplinary, theoretical, methodological aspects and the construction of scientific consensus on the contemporary phenomenon of information disorder. We also consider it important to extend the analysis to other databases to cover scientific production in local contexts.

While our study sheds light on the current landscape of misinformation research in the Social Sciences, it is essential to acknowledge certain limitations. Firstly, our analysis is primarily based on data from Web of Science and Scopus databases, and while these are reputable sources, they may not encompass all relevant publications in the field. Additionally, the scope of our study is confined to the period between 2017 and 2022, and the dynamic nature of research in this rapidly evolving field means that newer developments may not be fully captured. Furthermore, the focus on highly cited texts may introduce a potential bias towards well-established works, possibly overlooking emerging perspectives.

Despite the progress captured in this study, it is imperative to acknowledge certain areas for improvement and future exploration. This paper provides an overview of the current landscape and also contributes to the scientific dialogue by illuminating trends, influential actors, and collaborative networks. By doing so, we aim to enrich the existing body of knowledge within the Social Sciences and offer insights that may be valuable for scholars seeking to understand and address disinformation and fake news. By recognizing the limitations of our current analysis and embracing the potential for future exploration, we hope to stimulate a more comprehensive and nuanced understanding of information disorder, fostering ongoing dialogue and scholarly development within the Social Sciences.

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